GlucoTwin: Multimodal Clinical Digital Twin Architecture

2-Hour Continuous Glucose Spike Prediction & Counterfactual Intervention Simulation | Team Aethan · Amity University Mumbai

1. DUAL DATA STREAMS

Static EHR + Dynamic Wearables

STATIC EHR STREAM (200 Indian T2D Pts)
• Demographics, BMI, T2D Duration
• Labs: HbA1c, Fasting BG, Lipids, eGFR
• Polygenic Risk: TCF7L2, PPARG, GRS
• Regimen & Comorbidities

DYNAMIC WEARABLE / IoT STREAM
• 5-min Continuous Glucose (CGM)
• Heart Rate Variability (HRV RMSSD)
• Heart Rate (bpm) & Autonomic Stress
• Multi-Stage Sleep (Deep/REM/Light)
• Step Count & Activity Mode
• 1.728M Readings (30 Days / 200 Pts)

2. FEATURE PIPELINE

Zero-Leakage Patient Splits

Static EHR Encoding (53 Features)
ICMR Asian-Indian Phenotype, TG/HDL,
Polygenic Allele Dosage (TCF7L2/PPARG)

Temporal Kinetics (54 Features)
• 15m/30m/1h/2h Rolling Mean, Velocity, Accel
• Glycemic Volatility Index, MAG, HRV & Sleep

Patient-Partitioned Splitter
Train: 140 Pts | Val: 20 Pts | Test: 40 Pts
Zero patient overlap across splits
X_temporal: (N, 24, 54) | X_static: (N, 53)

3. MULTIMODAL AI ENSEMBLE

Calibrated 2-Hour Spike Forecaster

BRANCH A: PyTorch Attention-LSTM (0.60)
• 2-Layer LSTM (128 hidden) + Temporal Self-Attention
• Static EHR MLP (53 → 64 → 32) Late Fusion

BRANCH B: Tabular XGBoost (0.40)
• 485 Engineered Features (432 Temporal + 53 Static)
• Histogram GBDT with Cost-Sensitive Weighting

GLUCOTWIN ENSEMBLE (Held-Out 40 Test Pts)
ROC-AUC: 0.9925 | PR-AUC: 0.9965 | Spec: 96.2%
Predicts 2-Hour Glucose Excursion (>180 mg/dL)

4. DIGITAL TWIN ENGINE

Virtual Patient & Simulator

Twin State & Trajectory Engine
24-Step (2h) Kinetic Forecast +
90% Confidence Bounds

Composite Risk Calculator
ADA/RSSDI Clinical Tiers (0–100)
Low · Moderate · High · Critical

Counterfactual What-If Simulator
• Post-Meal Walk (0–45m, GLUT4)
• Meal Carb Load (0–75g)
• Scheduled Medication Dose

5. FASTAPI CLINICAL MICROSERVICE (localhost:8000)

Real-Time Inference, Attribution & Counterfactual Simulation

GET /api/patients/{id}/dashboard — Full EHR + Polygenic Profile, 6h Wearable Stream & Forecast

GET /api/patients/{id}/predict — Live PyTorch LSTM + XGBoost Inference, Feature Attribution & Attention

POST /api/patients/{id}/simulate — Parametric Counterfactual Simulation (walk_minutes, meal_carbs_g, med)

GET /api/patients & /api/stats/overview — Cohort Triage Queue & Population Glycemic Surveillance

DPDP Act 2023 & HIPAA Sandbox Compliant | 100% Synthetic Anonymized Indian T2D Cohort

6. NEXT.JS CLINICAL EHR WORKSTATION (localhost:3000)

Physician Decision-Support UI for Proactive T2D Care

Cohort Priority Triage Queue — Ranks 200 Patients by 2h Spike Risk with Search & Tier Filters

Static EHR & Polygenic Card — HbA1c, eGFR, Lipids, BP, Meds + TCF7L2 / PPARG / GRS Biomarkers

Synchronized Dual-Stream Chart — CGM + 2h Forecast (Top) & Wearable HRV + HR + Steps (Bottom)

Interactive What-If Sliders & Multimodal Attribution — Walk/Carb/Med Controls + Signed mg/dL Drivers

Clean Hospital EHR Light Theme | TypeScript + Tailwind CSS + Recharts | 37 Automated Unit Tests